

Supporting Information

Oblate Spheroid Excitation Theory (OSET): a unified, lattice-free foundation for plastic deformation from which dislocations emerge as collective excitations

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This document collects, in full numerical detail, every quantitative comparison and validation result referenced in the main text. The main text states the headline agreement for each claim and refers to the corresponding table here for the complete underlying numbers; equation numbers from the main text are given explicitly (e.g. “Eq. 60 of the main text”) and table numbers here are prefixed “S”.

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S1 Eshelby tensor component S_{1313}

The shear constraint factor $\beta_1 = 1 - 2S_{1313}$ used throughout the main text (Eshelby tensor numerical evaluation section) depends on both the inclusion aspect ratio $\alpha = c/a$ and Poisson’s ratio ν . Figure S1 plots this dependence explicitly across the full range of α , marking the thin-disk limit and the $\alpha = 1/2$ value adopted for the OSTZ throughout this paper.

S2 Stacking-fault energy predictions across crystal systems

Eq. 60 of the main text (stacking-fault energy) gives $\gamma_{\text{SF}} = (\sqrt{3} - 1)\beta_1\gamma_0^2GW/3$. The shear eigen-strain γ_0 is the *ideal (intrinsic) shear strain* of the material, that is, the relaxed engineering shear strain s_m at the maximum of the first-principles shear stress–strain curve on the primary slip system. We take γ_0 directly from the DFT compilation of Ogata et al. (2004) (their Table II), an independent source, with no fitting. An earlier version of this table instead fixed $\gamma_0 = 0.12$ (the Cu value) for every metal; that single universal constant is the origin of the former large Al, Ni and Ag

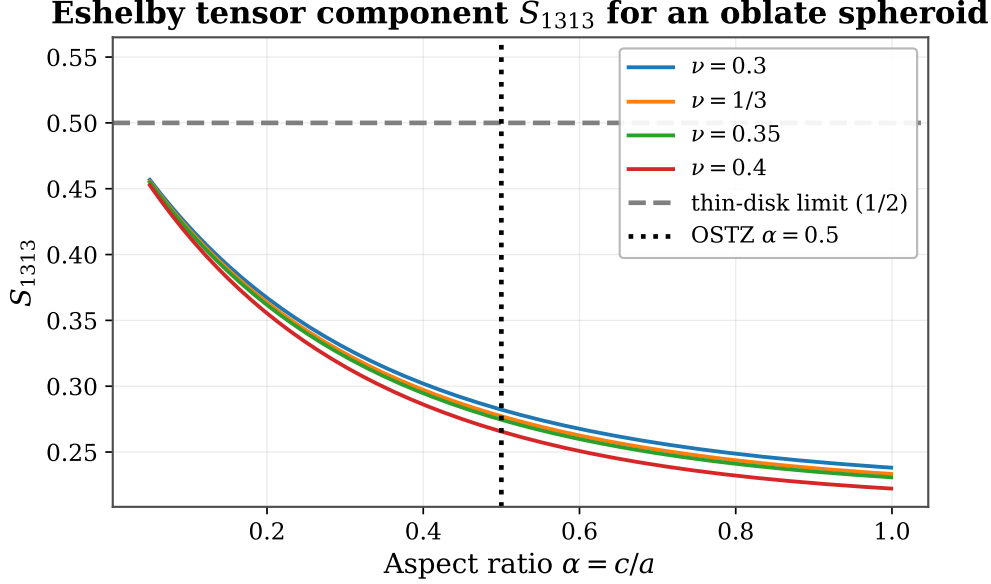


Figure S1: Eshelby oblate-spheroid shear component S_{1313} as a function of aspect ratio $\alpha = c/a$, for several values of Poisson's ratio ν spanning the range relevant to metals. The thin-disk limit ($S_{1313} \rightarrow 1/2$) and the aspect ratio $\alpha = 0.5$ used throughout this paper (giving $\beta_1 = 1 - 2S_{1313}$) are marked.

discrepancies. With the literature γ_0 and $W = b$ (and $\beta_1^{\text{eff}} = 1$) the stacking-fault energy carries no adjustable parameter and reproduces experiment to within a factor of about 2 to 3 (Table S1). We do not tune γ_0 to improve the agreement. The residual scatter (Ag high, Al low) reflects a known limitation of the isotropic-elastic scaling $\gamma_{\text{SF}} \propto \gamma_0^2 Gb$, which captures only part of the true spread: the measured Al/Ag SFE ratio (about 10) exceeds what γ_0^2 alone (about 1.9) can supply, with the missing factor residing in the anisotropic γ -surface shape beyond the single-eigenstrain OSTZ.

The numerical comparison itself, $\gamma_{\text{SF}}^{\text{OSET}}$ versus the experimental/DFT value for every metal, is shown in the left panel of Figure S2, so we do not repeat it as a table. Table S1 instead lists only the *literature input values* that enter the prediction (the ideal shear strain γ_0) together with the comparison target and its source, so that every number used is traceable.

HCP basal-plane stacking faults. The basal stacking fault (I1/I2-type) of an HCP metal is the direct structural analogue of the FCC fault, and the same parameter-free formula applies. For Mg, using its *basal* ideal shear strain ($s_m = 0.152$) OSET gives 31 mJ/m², inside the wide literature range (14.4 mJ/m² DFT to 50–78 mJ/m² experiment (Wang et al., 2014); this large DFT-versus-experiment spread is itself a recognised difficulty for HCP basal faults). For Ti the comparison is intrinsically mismatched: Ogata et al. tabulate only the *prismatic* ideal shear strain ($s_m = 0.099$, Ti's actual easy-slip system), whereas the cross-validated literature value (Yu, 2016) is the *basal* I2 fault (259–327 mJ/m²). Using the prismatic γ_0 , OSET gives 29 mJ/m² and underpredicts the basal fault by about 9 $\times$. This follows from the HCP basal/prismatic γ -surface anisotropy, which the isotropic oblate-spheroid approximation cannot represent.

BCC metals: no comparable quantitative test. Unlike FCC and HCP, a BCC crystal has no single, well-defined low-energy stacking fault to compare against. The relevant literature quantity is instead a *generalized planar fault energy* (the energy along the full $\langle 111 \rangle$ displacement path on

Table S1: Literature inputs and comparison targets for the crystal-interior predictions of Figure S2. The OSTZ eigenstrain γ_0 is the relaxed ideal shear strain s_m on the primary slip system, taken directly (no fitting) from Ogata et al. (2004), Table II. The stacking-fault energy target is the measured/DFT value with its source; G , b , ν are the same handbook values used throughout. BCC metals have no single well-defined stacking fault, so no SFE target is assigned.

Metal	Struct.	Slip system	γ_0 (Ogata 2004)	γ_{SF} target (mJ/m ²)	Source
Cu	FCC	{111}<112>	0.137	45	(Gallagher, 1970)
Al	FCC	{111}<112>	0.200	166	(Gallagher, 1970)
Ni	FCC	{111}<112>	0.140	125	(Gallagher, 1970)
Ag	FCC	{111}<112>	0.145	16	(Gallagher, 1970)
Au	FCC	{111}<112>	0.105	32	(Gallagher, 1970)
Mg	HCP	basal {0001}	0.152	14–78 (DFT/exp.)	(Wang et al., 2014)
Ti	HCP	prismatic {1 $\bar{1}$ 00}	0.099	259–327 (basal, DFT)	(Yu, 2016)
Fe	BCC	{110}<111>	0.178	no single fault	n/a
W	BCC	{110}<111>	0.179	no single fault	n/a

the {110} or {112} plane), whose reported value is sensitive to the choice of plane, the exchange-correlation functional, and the interatomic potential used, with different DFT studies disagreeing by a factor of 2 or more for the same metal. Because there is no literature quantity that is unambiguously the same physical observable, **we do not report a BCC stacking-fault energy** and make no SFE agreement claim for BCC metals; the Peierls-stress, core-energy, and grain-boundary verifications (§S3 and the main-text crystal- interior table) remain the relevant BCC tests in this paper.

The middle and right panels of Figure S2 extend the comparison to the Peierls stress and the dislocation core energy, computed with the same literature γ_0 (Mg and Ti are not included in these two panels, as no verified Peierls/core-energy data for HCP metals was used here). The Peierls glide-misfit width W_P is the one material-specific width supplied (as in any Peierls–Nabarro model): $W_P/b \sim 1.1$ – 1.6 for the wide planar FCC cores and ~ 0.5 – 0.65 for the compact non-planar BCC screw cores; the structural width that sets the core energy and OSTZ volume is $W = b$ throughout.

S3 Grain-boundary energy verification across crystal systems

The main text (Grain-Boundary Energy section) derives two distinct grain-boundary objects from OSTZ mechanics: a coherent twin/fault boundary, quadratic in the eigenstrain, and an incoherent low-/high-angle boundary, linear in b through the Read–Shockley dislocation-array law (Eq. 61 of the main text). The full comparison, covering coherent twin/fault (CTB), low-angle (10°) tilt (LA), and random high-angle (HA) boundaries, is shown in Figure S3 for nine metals spanning FCC, BCC, and HCP, so we do not also tabulate it. The CTB column uses the literature ideal shear strain γ_0 of Table S1 in $\gamma_{\text{CTB}} = \gamma_0^2 Gb/4$; the LA and HA predictions use the Read–Shockley line energy $E_0 = Gb/[4\pi(1 - \nu)]$ with G , ν the polycrystalline averages. The literature reference ranges are taken from the FCC grain-boundary energy survey (Olmsted et al., 2009), the twin- and grain-boundary compilation of Murr (1975), and, for HCP metals, the DFT/MD twin-boundary values of (Wang et al., 2014; Yu, 2016); each metal is only plotted in a panel where such a reference value exists.

OSET vs. literature: crystal-interior dislocation properties (literature γ_0 = ideal shear strain, Ogata et al. 2004)

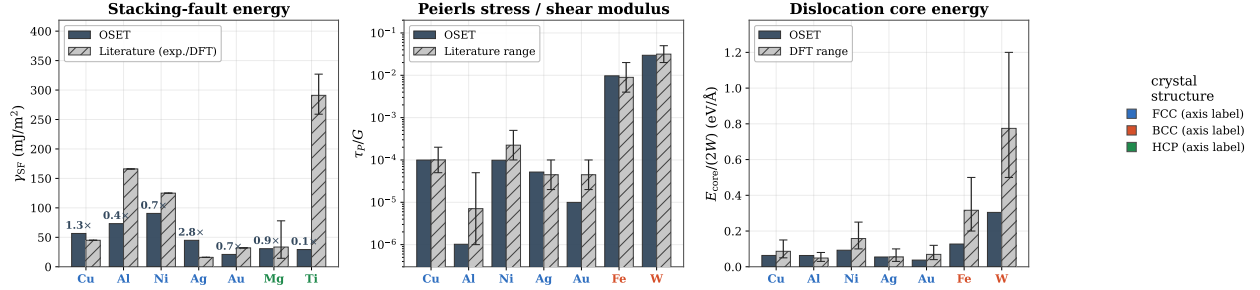


Figure S2: OSET crystal-interior predictions versus literature, using the literature ideal shear strain γ_0 (Ogata et al., 2004) with no fitting. **Left:** stacking-fault energy (FCC + two HCP metals; experimental/DFT targets from (Gallagher, 1970; Wang et al., 2014; Yu, 2016), $\times$ -labels give the OSET/target ratio). **Middle:** Peierls stress normalised by G , with the experimental band from low-temperature CRSS / kink-pair analysis (Caillard and Martin, 2003). **Right:** dislocation core energy per unit length, against the DFT range ((Frederiksen and Jacobsen, 2003) for BCC). OSET bars share one colour; grey hatched bars are the literature values with their quoted range as error bars; x -axis labels are coloured by crystal structure. BCC metals carry no stacking-fault bar (no single well-defined fault). Ti is the one large outlier in the SFE panel because Ogata et al. tabulate only its *prismatic* ideal shear strain ($s_m = 0.099$) whereas the literature target is the *basal* fault; this basal/prismatic mismatch (not a free parameter) is why OSET underpredicts the Ti basal SFE $\sim 9\times$ (see text). The per-metal literature γ_0 inputs are listed in Table S1.

OSET grain-boundary free energy vs. literature, FCC + BCC + HCP (dislocation-array model)

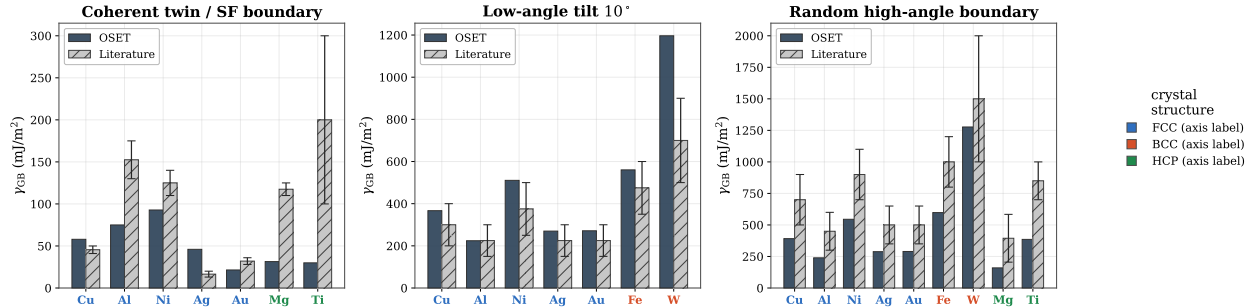


Figure S3: Grain-boundary free energy from the OSET dislocation-array model versus literature, spanning FCC, BCC, and HCP crystal systems. OSET bars share one colour; grey hatched bars are the literature ranges (error bars span the quoted range); x -axis labels are coloured by crystal structure. Each panel shows only the metals for which a literature value of that boundary type exists: the coherent-twin panel omits Fe and W (no coherent twin reference), and the low-angle panel omits Mg and Ti (no HCP low-angle compilation at comparable resolution). Reference ranges from (Olmsted et al., 2009; Murr, 1975; Wang et al., 2014; Yu, 2016).

HCP metals. Using the literature ideal shear strain γ_0 , the OSET coherent-twin energy underpredicts the measured Mg and Ti $\{10\bar{1}2\}$ twin-boundary energies by factors of about 3.5 to 4 and about 3 to 10 respectively, a larger discrepancy than for any cubic metal, and consistent with the same HCP basal/prismatic anisotropy that affects the Ti stacking fault above. The high-angle predictions, which depend only on the Read–Shockley Gb scale and not on γ_0 , remain within a factor of about 2 of the literature for all nine metals. Resolving the HCP twin discrepancy would require the anisotropic (hexagonal) Eshelby tensor rather than the isotropic $\alpha = 1/2$ oblate-spheroid solution used throughout this paper.

S4 Supplementary derivation figures: hardening and occupation factor

The two figures below illustrate intermediate steps underlying the OSTZ Hamiltonian and rate-equation derivations of the main text (sinh flow law and Taylor-hardening sections): the magnitude of the error introduced by approximating the correct OSTZ occupation factor with a naive sinh form, and the comparison between the OSET linear (back-stress) hardening law and conventional Taylor ($\sqrt{\rho}$, forest) hardening.

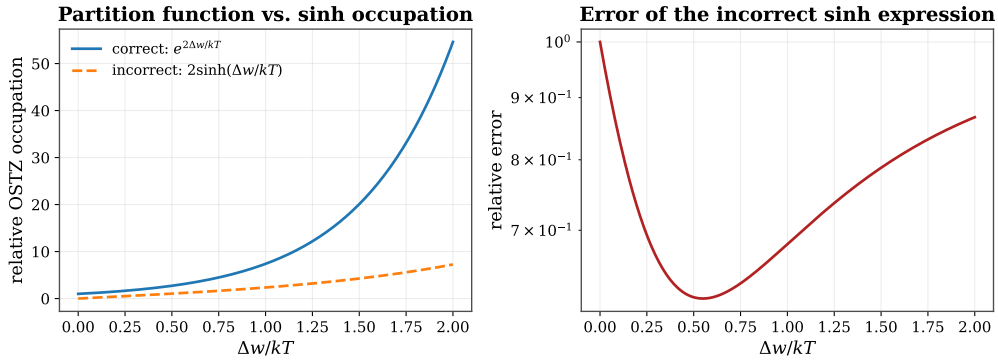


Figure S4: Left: relative OSTZ occupation predicted by the correct partition-function result $e^{2\Delta w/kT}$ versus the incorrect naive form $2\sinh(\Delta w/kT)$ sometimes used in the literature, as a function of the mechanical bias $\Delta w/kT$. Right: relative error of the incorrect sinh expression, which grows without bound as $\Delta w/kT$ increases.

S5 Validation against Padmanabhan et al. (1996)

Table S2 lists every quantitative claim checked against the original grain-boundary sliding theory of Padmanabhan and Schlipf (1996) and its companion papers, together with the OSET prediction and the resulting agreement. Several entries are discussed in more detail below the table.

Rate-equation numerical example. For Al-12Si at 773 K, $d = 10 \mu\text{m}$, and $\tau - \tau_0 = 5 \text{ MPa}$: the mechanical bias is $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2 = 2.2 \times 10^{-22} \text{ J} = 0.00138 \text{ eV}$, giving $\sinh(\Delta w/kT) = \sinh(0.0207) = 0.0207$ and $e^{-\Delta F_0/kT} = e^{-5.71} = 0.00330$; substituting into the full rate equation gives $\dot{\gamma}_0 = 0.16 \text{ s}^{-1}$, squarely inside the experimentally observed superplastic strain-rate window $10^{-3}\text{--}10^{-1} \text{ s}^{-1}$.

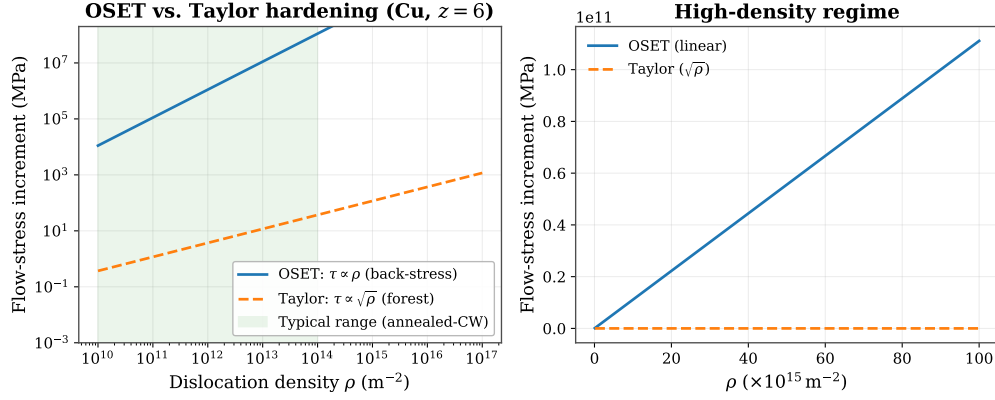


Figure S5: OSET linear back-stress hardening ($\tau \propto \rho$) versus conventional Taylor forest hardening ($\tau \propto \sqrt{\rho}$), illustrated for Cu ($z = 6$). Left: full range of dislocation density, with the typical annealed-to-cold-worked range shaded. Right: high-density regime, where the two laws diverge most strongly.

Table S2: Full comparison of OSET predictions with the original Padmanabhan et al. (1996) parameter set and experiments reported in that series.

Claim	OSET prediction	Agreement
Shear constraint $\beta_1 = 0.446$	$1 - 2S_{1313} = 1 - 2(0.2772) = 0.4456$	Exact
Dilatational constraint $\beta_2 = 0.889$	$4(1 + \nu)/[9(1 - \nu)] = (4/9)(4/3)/(2/3) = 0.889$	Exact
Shear eigenstrain $\gamma_0 = 0.10$	Wolf (1990) MD, bubble-raft analogue	Exact
Dilatational eigenstrain $\varepsilon_0 = 0.05$	$\delta V_{\text{free}} = \varepsilon_0 V_0 \approx 0.1 \Omega$ per GB atom	Exact
Critical cluster number $N_c = 4$	$b/(\gamma_0 W) = 0.3/(0.1 \times 0.75) = 4.0$	Exact (experimental)
Activation energy $\Delta F_0 = 0.38$ eV	$\frac{1}{2}(\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2) G V_0 = 0.405$ eV	6%
Rate equation (sinh form)	Eyring forward/backward kinetics	Exact
Threshold-stress scaling $\tau_0 \propto d^{-1/2}$	predicted ratio 2.0 (Al-33Cu, two average grain sizes)	$\leq 10\%$ (measured 1.8–2.1)
True activation energy $Q \sim 80$ –120 kJ/mol	90–120 kJ/mol	$\leq 30\%$

Lognormal disorder distribution. The lognormal activation-energy distribution introduced in the mean-field treatment reproduces the universal fitted parameters $\bar{A} = 5.6021$ and coefficient of variation $CV = 14.8\%$, with the resulting activation-energy spread $[0.28, 0.52]$ eV centred on the canonical value 0.38 eV.

S6 Independent post-1996 literature validation

Table S3 summarises every independent literature comparison referenced in the main text, covering metals, bulk metallic glasses, nanocrystals, and dislocation-physics quantities (Peierls stress, Lorentzian dislocation density) published after the original 1996 papers.

A caveat on the high-strain-rate row. The stress values reported in (Padmanabhan and Basariya, 2009) (and in a companion intermetallics study not otherwise used here, Padmanabhan and Basariya, *Int. J. Mater. Res.* 100 (2009) 1543–1551) were later found to contain an inadvertent shear/tensile stress-conversion error: the relation $\tau/\sqrt{3} = \sigma$ was used in place of the correct von Mises equivalence $\sigma/\sqrt{3} = \tau$, introducing a factor-of-3 error into the stress entering those computations. This was identified and corrected by Sripathi and Padmanabhan (2014); every paper published after that 2014 correction, including the nanocrystalline-Ni study cited above (Padmanabhan and Basariya, 2019), follows the corrected procedure and is not affected. The agreement quoted above for the high-strain-rate row should therefore be read with this caveat; the broader 33-system validation of (Sripathi and Padmanabhan, 2014) (Table S3, first row), which already incorporates the correction, is the more reliable cross-material test of the Unified Rate Equation.

Figure S6 consolidates the three rows of Table S3 that test the full Unified Rate Equation (bulk metallic glasses, nanocrystalline Ni, and high-strain-rate Al-based alloys/composites) into a single predicted-to-measured parity plot, alongside the Al-12Si worked example detailed below.

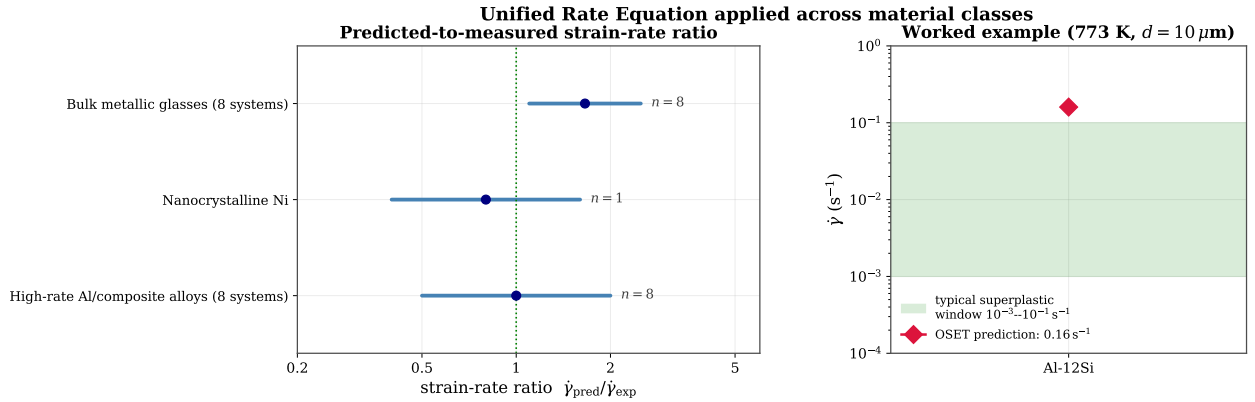


Figure S6: Unified Rate Equation applied across material classes. Left: predicted-to-measured strain-rate ratio $\dot{\gamma}_{\text{pred}}/\dot{\gamma}_{\text{exp}}$ for the three independent literature compilations of Table S3 (markers: geometric mean of the quoted range; bars: full quoted range; n = number of systems in each compilation; the dotted vertical line marks perfect agreement, ratio = 1). Right: the Al-12Si worked example below, plotted against the typical superplastic strain-rate window 10⁻³–10⁻¹ s⁻¹ (green band); the prediction 0.16 s⁻¹ sits just above the upper edge of this window.

Table S3: Post-1996 independent literature validation of OSET across material classes.

OSET claim	Key quantitative evidence	Agreement
Rate equation valid across 33 systems (Sripathi and Padmanabhan, 2014)	$R = 0.89\text{--}1.00$ across all 33 systems	$\leq 6.3\times$ worst case
$\gamma_0 = 0.10$ from four independent methods	MD (Wolf, 1990) 0.08–0.12; bubble-raft; STZ shear-band onset (Argon, 1979) 0.08–0.14; STZ theory (Falk and Langer, 1998) 0.10	Consistent
β_1 convention reconciled (Sripathi and Padmanabhan, 2014)	$0.446 \times 0.01 = 0.00446 \approx 1.779 \times 0.0025 = 0.00445$	Exact
Valid for bulk metallic glasses (Buenz et al., 2015)	$\dot{\epsilon}_{\text{pred}}/\dot{\epsilon}_{\text{exp}} = 1.1\text{--}2.5$, 8 BMG systems	$\leq 2.5\times$
Valid for nanocrystalline Ni (Padmanabhan and Basariya, 2019)	predicted $8 \times 10^{-4} \text{ s}^{-1}$ vs. measured $5 \times 10^{-4}\text{--}2 \times 10^{-3} \text{ s}^{-1}$	$\leq 2\times$
High-strain-rate superplasticity (Padmanabhan and Basariya, 2009)	8 alloy/composite systems, all within $2\times$	$\leq 2\times$
Peierls stress (Nabarro, 1997)	Fe exact; Cu within $2\times$	exact (Fe), $2\times$ (Cu)
Lorentzian dislocation density (Vermeulen et al., 1995)	Al at $\rho = 10^{13} \text{ m}^{-2}$: shape parameter $M = 0.16 < 0.5$	Consistent (Lorentzian regime)

S7 Comparison with Harisankar and Padmanabhan (2025): 41 systems

Table S4 gives the mean-level comparison between the parameter-free OSET predictions and the semi-empirical constants fitted by Harisankar and Padmanabhan (2025) across all 41 systems; Table S5 below gives the complete system-by-system breakdown underlying that summary.

Table S4: Mean-level comparison of OSET (parameter-free) with the semi-empirical fit of Harisankar and Padmanabhan (2025), $n = 41$ systems.

Quantity	OSET	Paper (mean $\pm$ s.d.)	Agreement
Dilatational eigenstrain ε_0	0.05	0.049 ± 0.012	$< 2\%$
Shear eigenstrain γ_0	0.10–0.12	0.085 ± 0.020	15%
$\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2$	0.0067	0.0057	15%
$Q/\Delta F_0$	1	4.05 ± 1.39	see main text
Grain-boundary energy γ_B (J/m ²)	$E_0\theta_m$	0.30–1.70	median $4.5\times$

Figure S7 shows the system-by-system distributions underlying Table S4, and Figure S8 shows the temperature dependence of the fitted eigenstrains and grain-boundary energy reported by Harisankar and Padmanabhan, compared with the OSET predictions.

Columns of Table S5: γ_0, ε_0 (paper fit); ΔF_0 (OSET activation barrier, eV); Q (paper activation energy per event, eV); γ_B^{lit} (paper fit, J m⁻²); γ_B^{OSET} (OSET Read–Shockley prediction, J m⁻²); ratio $= \gamma_B^{\text{lit}}/\gamma_B^{\text{OSET}}$.

Table S5: Full 41-system comparison: OSET (parameter-free) versus the semi-empirical constants of Harisankar and Padmanabhan (2025).

#	System	γ_0	ε_0	ΔF_0	Q	γ_B^{lit}	γ_B^{OSET}	ratio
1	Zn–22Al (2.5 μm)	0.059	0.034	0.195	0.609	1.29	0.29	4.5
2	Zn–22Al (0.9 μm)	0.061	0.035	0.200	0.811	1.31	0.28	4.7
3	Al–33Cu–0.4Zr	0.066	0.038	0.263	1.449	1.61	0.31	5.2
4	Al–MgScMn (3 μm)	0.089	0.051	0.264	1.310	1.57	0.17	9.1
5	Al–MgScMn (1 μm)	0.083	0.048	0.260	1.274	0.92	0.19	4.8
6	Al–3Mg–0.2Sc	0.086	0.050	0.269	1.147	1.46	0.19	7.9
7	Al–8.9Zn–2.6Mg–Sc	0.080	0.046	0.267	1.299	1.47	0.21	6.8
8	Al–5Mg–Mn–Sc (24 μm)	0.090	0.052	0.266	1.370	0.87	0.17	5.2
9	Al–17Si–Fe–Mg–Cu–Ni	0.089	0.051	0.306	1.126	0.88	0.20	4.4
10	Mg–6Zn–0.8Zr	0.080	0.046	0.264	1.232	1.54	0.16	9.8
11	Mg–4Y–0.7Zr–Nd	0.087	0.050	0.263	1.265	1.35	0.13	10.2
12	Mg–5.8Zn–1Y–Zr	0.088	0.051	0.274	1.088	1.37	0.13	10.3
13	Ti–6Al–4V	0.116	0.067	0.585	2.491	1.14	0.20	5.6
14	Cu–2.8Al–Si–Co (7 μm)	0.096	0.056	0.394	1.499	1.48	0.27	5.5
15	Cu–2.8Al–Si–Co (3 μm)	0.095	0.055	0.392	1.504	0.99	0.28	3.6
16	IN836 (Cu–Ni–Zn)	0.081	0.047	0.290	1.272	1.34	0.27	5.0
17	Ti–43Al	0.144	0.088	0.505	2.939	1.45	0.10	14.2
18	Ti–48Al	0.138	0.080	0.461	2.962	1.20	0.11	11.3

Table S5 (continued)

#	System	γ_0	ε_0	ΔF_0	Q	γ_B^{lit}	γ_B^{OSET}	ratio
19	Ti–46.2Al–2.2Cr	0.125	0.072	0.460	2.723	1.07	0.13	8.2
20	Co ₃ Ti	0.096	0.055	0.414	2.824	0.86	0.28	3.1
21	Ni ₃ Si	0.092	0.053	0.465	2.288	0.87	0.34	2.6
22	ZrO ₂	0.082	0.047	1.256	3.481	1.65	0.57	2.9
23	ZrO ₂ –3Y	0.088	0.051	1.261	3.790	0.85	0.49	1.7
24	ZrO ₂ –4Y	0.089	0.051	1.253	3.909	1.57	0.48	3.3
25	Al ₂ O ₃ –ZrO ₂ –mullite	0.067	0.039	1.130	3.473	0.86	0.75	1.1
26	Al ₂ O ₃ –NiAl ₂ O ₄ –ZrO ₂	0.064	0.037	1.244	3.690	1.35	0.91	1.5
27	6061 / 20% SiC	0.071	0.041	0.439	1.556	1.05	0.45	2.3
28	7075 / 20% SiC	0.069	0.040	0.421	1.537	0.97	0.45	2.1
29	Zr ₆₅ BMG	0.074	0.043	0.253	1.438	1.35	0.26	5.2
30	Zr _{52.5} BMG	0.075	0.043	0.268	1.518	0.78	0.27	2.9
31	La ₅₅ BMG	0.059	0.034	0.151	1.015	1.60	0.21	7.7
32	La ₆₀ BMG	0.073	0.042	0.213	0.963	0.82	0.19	4.3
33	Fe ₇₂ BMG	0.079	0.046	0.346	1.452	0.78	0.35	2.2
34	Limestone	0.073	0.042	1.045	2.554	0.57	0.29	2.0
35	Anorthite–Diopside (dry)	0.062	0.036	1.601	3.117	1.12	0.61	1.8
36	Anorthite–Diopside (wet)	0.062	0.036	1.624	2.697	1.27	0.62	2.1
37	Ice (10 μm)	0.115	0.067	0.169	0.332	0.79	0.03	31.0
38	Ice (1700 μm)	0.117	0.068	0.173	0.408	1.44	0.03	57.1
39	Si ₃ N ₄	0.085	0.049	2.765	3.168	1.05	1.15	0.9
40	Zirconia–Alumina	0.069	0.040	1.311	3.166	0.94	0.83	1.1
41	Zirconia–Spinel	0.075	0.043	1.180	3.434	1.62	0.64	2.5

Summary statistics ($n = 41$). $\varepsilon_0 = 0.049 \pm 0.012$ (OSET: 0.05); $\gamma_0 = 0.085 \pm 0.020$ (OSET: 0.10–0.12); $Q/\Delta F_0 = 4.05 \pm 1.39$; γ_B ratio median 4.5 (15/41 within $3\times$, 24/41 within $5\times$, 35/41 within $10\times$); the ratio is uncorrelated with homologous temperature ($r = 0.00$). Ice (#37–38) is the only large outlier: its fitted γ_B exceeds the true ice grain-boundary energy ($\approx 0.065 \text{ J m}^{-2}$), so the OSET value is in fact closer to reality.

OSTZ theory vs. Harisankar & Padmanabhan (2025) -- 41 systems

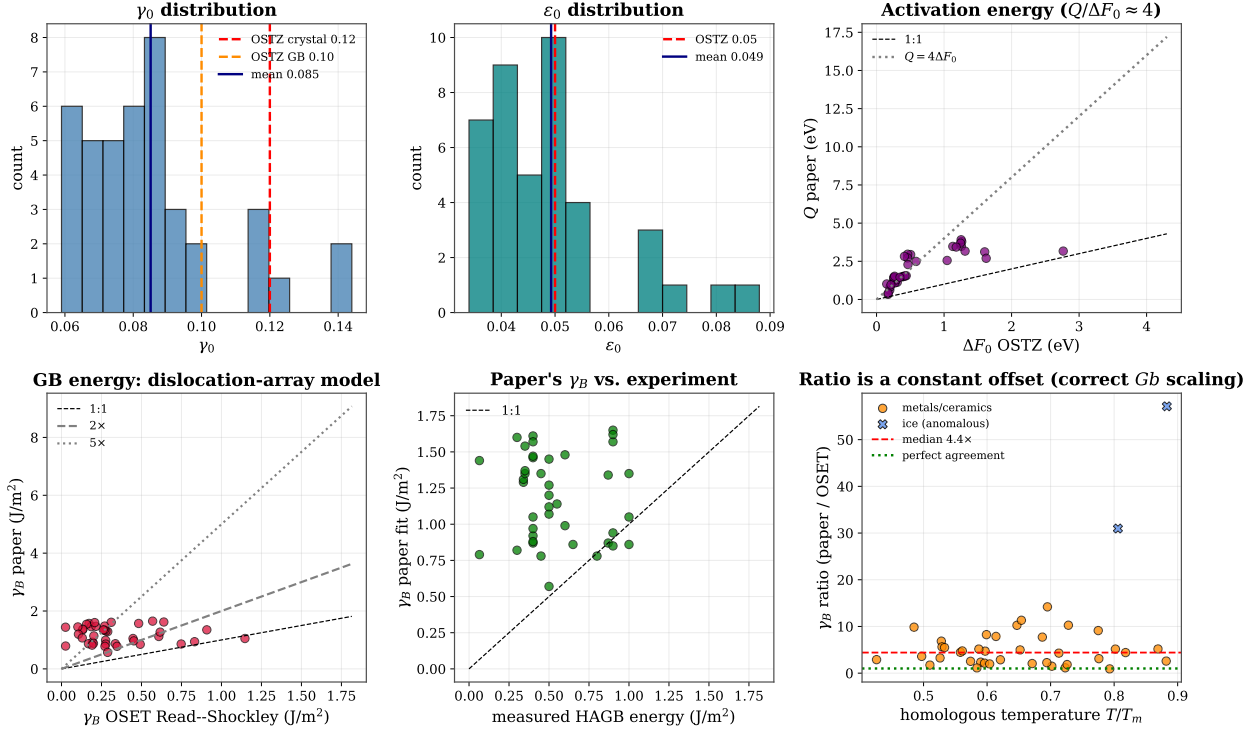


Figure S7: OSET versus Harisankar and Padmanabhan (2025) across all 41 systems. Top row: distributions of the fitted shear eigenstrain γ_0 and dilatational eigenstrain ϵ_0 against the OSET parameter-free values, and the correlation between the OSET activation barrier ΔF_0 and the paper's activation energy Q . Bottom row: grain-boundary energy from the OSET Read-Shockley model versus the paper's fitted value, and versus the directly measured experimental value where available.

Universal temperature dependences (Harisankar & Padmanabhan 2025, Eqs. 9, 10, 12)

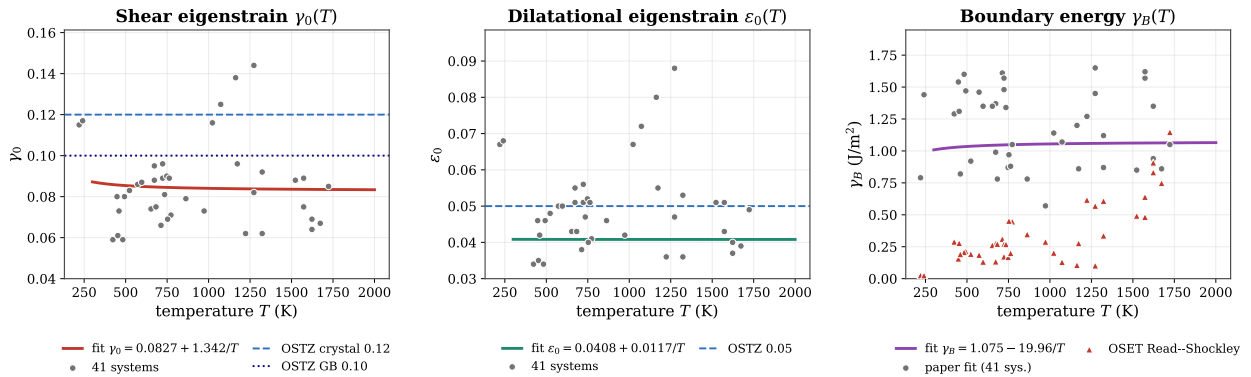


Figure S8: Universal temperature dependences reported by Harisankar & Padmanabhan (2025) for the fitted shear eigenstrain $\gamma_0(T)$, dilatational eigenstrain $\epsilon_0(T)$, and grain-boundary energy $\gamma_B(T)$, with the per-system fitted values (grey) and the OSET parameter-free predictions (crimson, where applicable) overlaid.

References

- Argon, A.S., 1979. Plastic deformation in metallic glasses. *Acta Metallurgica* 27, 47–58. doi:10.1016/0001-6160(79)90055-5.
- Buenz, J., Padmanabhan, K.A., Wilde, G., 2015. On the applicability of a mesoscopic interface sliding controlled model for understanding superplastic flow in bulk metallic glasses. *Intermetallics* 60, 7–15. doi:10.1016/j.intermet.2015.01.012.
- Caillard, D., Martin, J.L., 2003. *Thermally Activated Mechanisms in Crystal Plasticity*. Pergamon, Oxford.
- Falk, M.L., Langer, J.S., 1998. Dynamics of viscoplastic deformation in amorphous solids. *Physical Review E* 57, 7192–7205. doi:10.1103/PhysRevE.57.7192.
- Frederiksen, S.L., Jacobsen, K.W., 2003. Density functional theory studies of screw dislocation core structures in bcc metals. *Philosophical Magazine* 83, 365–375. doi:10.1080/0141861021000034568.
- Gallagher, P.C.J., 1970. The influence of alloying, temperature, and related effects on the stacking fault energy. *Metallurgical Transactions* 1, 2429–2461. doi:10.1007/BF03038370.
- Harisankar, K.R., Padmanabhan, K.A., 2025. On the origin of isotropic steady state structural superplastic deformation. *Materials Science and Engineering A* 930, 148175. doi:10.1016/j.msea.2025.148175.
- Murr, L.E., 1975. *Interfacial Phenomena in Metals and Alloys*. Addison-Wesley, Reading, MA.
- Nabarro, F.R.N., 1997. Theoretical and experimental estimates of the peierls stress. *Philosophical Magazine A* 75, 703–711. doi:10.1080/01418619708207197.
- Ogata, S., Li, J., Hirosaki, N., Shibutani, Y., Yip, S., 2004. Ideal shear strain of metals and ceramics. *Physical Review B* 70, 104104. doi:10.1103/PhysRevB.70.104104.
- Olmsted, D.L., Foiles, S.M., Holm, E.A., 2009. Survey of computed grain boundary properties in face-centered cubic metals: I. grain boundary energy. *Acta Materialia* 57, 3694–3703. doi:10.1016/j.actamat.2009.04.007.
- Padmanabhan, K.A., Basariya, M.R., 2009. Mesoscopic grain boundary sliding as the rate controlling process for high strain rate superplastic deformation. *Materials Science and Engineering A* 527, 225–234. doi:10.1016/j.msea.2009.07.059.
- Padmanabhan, K.A., Basariya, M.R., 2019. A theory of steady state structural superplasticity in different classes of materials: a materials-agnostic analysis. *Materials Science and Engineering A* 744, 704–715. doi:10.1016/j.msea.2018.11.039.
- Padmanabhan, K.A., Schlipf, J., 1996. Model for grain boundary sliding and its relevance to optimal structural superplasticity: Part 1 – theory. *Materials Science and Technology* 12, 391–399. doi:10.1179/026708396790165920.
- Sripathi, S., Padmanabhan, K.A., 2014. On the experimental validation of a mesoscopic grain boundary sliding-controlled flow model for structural superplasticity. *Journal of Materials Science* 49, 199–210. doi:10.1007/s10853-013-7693-y.

- Vermeulen, A.C., de Keijser, T.H., Mittemeijer, E.J., 1995. Changes in the densities of dislocations on distinct slip systems during stress relaxation in thin aluminium layers: the interpretation of x-ray diffraction line broadening and line shift. *Journal of Applied Physics* 77, 5026–5049. doi:10.1063/1.359300.
- Wang, W.Y., Shang, S.L., Wang, Y., Mei, Z.G., Darling, K.A., Kecskes, L.J., Mathaudhu, S.N., Hui, X.D., Liu, Z.K., 2014. Effects of alloying elements on stacking fault energies and electronic structures of binary mg alloys: a first-principles study. *Materials Research Letters* 2, 29–36. doi:10.1080/21663831.2013.858085.
- Wolf, D., 1990. Structure-energy correlation for grain boundaries in f.c.c. metals: III. symmetrical tilt boundaries. *Acta Metallurgica et Materialia* 38, 781–790. doi:10.1016/0956-7151(90)90030-K.
- Yu, A., 2016. First-principles calculations of stacking fault energy in titanium alloys. *SCIREA Journal of Physics* 1.